

Zero-Discharge Bioleaching of Spent Lithium-Ion Black Mass: Process Design and the Cost of a Biological Lixiviant

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In the 21st century, the rapid advancement of AI and new technologies has caused global energy consumption to skyrocket. Electrical use is now central to every industry and lifestyle, bringing to light the harsh limitations of our finite planet. As we are taught, matter is neither created nor destroyed, only transformed, yet our reserves of critical metals are dwindling at an unsustainable rate. Projections indicate that lithium-ion battery (LIB) demand will outpace raw material supply by 2030 [1], with nearly 70% of global cobalt reserves consumed for batteries by 2040 [2]. These cobalt shortages are driven not just by rising demand, but also by the acute geographic concentration of reserves in the Democratic Republic of Congo (DRC), which exposes the supply chain to significant geopolitical and operational risks. Meeting this massive demand for clean energy will require increasing lithium supplies by 18-20 times and cobalt by 17-19 times by 2050 [3]. To bridge this gap, we must pivot toward sustainable recovery methods like microbial bioleaching, which can recover battery-grade metals while reducing CO₂ emissions to just 16-19 kg per kg of cobalt, a significant improvement over the 43-91 kg footprint associated with traditional acid leaching [4].

Conventional critical material recovery relies on direct recycling, pyrometallurgy, and hydrometallurgy, each presenting distinct technological and environmental constraints. Direct recycling preserves the original crystal structure of cathode materials to minimize processing energy and chemical consumption, yet it requires precise, complex pre-sorting of cathode chemistries. Pyrometallurgy utilizes high-temperature smelting to process diverse cathode types, but it is extremely energy-intensive, generates hazardous emissions, and permanently loses volatile elements like lithium and aluminum to the slag. Hydrometallurgy uses inorganic acids and reducing agents to extract soluble metal salts at high yields, but it demands immense chemical and water inputs alongside extensive wastewater treatment infrastructure. Although these methods are commercially established, their scalability in the United States is constrained by strict environmental regulations, high capital costs, and substantial secondary waste burdens, underscoring the urgent need for a cleaner, sustainable bioleaching alternative.

In "Sustainable bioleaching of lithium-ion batteries for critical materials recovery" [4], the authors evaluate a biolixiviant to leach critical metals out of cathode-containing powder, or black mass. The lixiviant is a gluconic-acid broth generated by *Gluconobacter oxydans* grown on corn stover hydrolysate, and the leach is run in the presence of Iron(II), added as ferrous sulfate, to promote metal dissolution. Although the process is projected to operate at an average profit

margin of roughly 21% on a basis of 10,000t of black mass per year, it would still require significant waste processing. The bulk of that processing burden traces back to the iron itself. Because Fe(II) is not a catalyst but a stoichiometric reductant (oxidized to Fe(III) as it converts Co(III) to the soluble Co(II) state) essentially all of the 3.75–4.51 kg of iron sulfate consumed per kilogram of recovered cobalt reports to the pregnant leach solution as ferric iron. Iron cannot be neutralized concurrently with target metals; Fe(III) hydrolyzes near pH 3–3.5, whereas Fe(II), Co(II), and Ni(II) precipitate near pH 8. Consequently, any residual ferrous iron must be fully oxidized and removed prior to metal recovery to prevent significant cobalt losses to the iron sludge. The resulting ferric hydroxide is bulky, difficult to filter, and occludes some cobalt and nickel as it settles. Complicating this further, gluconic acid is used industrially as an iron sequestrant, so the same lixiviant that enables the leach may hold Fe(III) in solution and force an upstream organics-destruction step before the iron will precipitate cleanly.

Beyond the complexities of iron management, the spent leach liquor retains a significant organic waste load, generating approximately 183 to 216 kg of wastewater per kilogram of recovered cobalt consisting of residual gluconate, fermentation metabolic byproducts, and unconsumed sugars. While this high chemical oxygen demand (COD) effluent represents a substantial treatment load, its inherent biodegradability highlights the core environmental advantage of organic bioleaching compared to persistent mineral-acid raffinate. Sulfate ions introduced through ferrous sulfate additions also require rigorous handling, typically requiring precipitation as solid gypsum during lime neutralization and generating a secondary bulk solid waste stream. Concurrently, the bioprocess media introduces regulated nutrient species, as both the phosphate salts and nitrogen sources within the growth medium necessitate dedicated nutrient removal steps. Prior to solvent extraction circuits, residual copper and aluminum contaminants originating from current-collector foil fines must be systematically eliminated; copper interferes competitively with target metal extraction, while aluminum forms gelatinous hydroxide precipitates that severely impair filtration performance. Fluoride species originating from residual LiPF_6 salt and PVDF binder are conventionally precipitated as insoluble calcium fluoride. Finally, even at respectable baseline extraction efficiencies of 86% cobalt and 84% nickel, the unreacted solid residue (comprising residual graphite, binder, and unleached cathode fragments) retains approximately one-seventh of the initial cobalt content—representing both a solid waste disposal liability and the single largest yield loss in the process. Taken together, these interconnected waste streams delineate a clear but nuanced process balance: while the effluent offers superior biodegradability compared to inorganic acid waste, realizing this ecological benefit relies on specialized iron, fluoride, and nutrient purification trains that present engineering and capital challenges rather than insurmountable chemical barriers. Consequently, the commercial viability of this bioleaching pathway ultimately depends less on the primary leach reaction itself than on how effectively these remediation circuits are integrated around it, a central question addressed by the process flow diagram developed over the course of this summer's research.

The developed process flow diagram models an annual feed rate of 10,000 tonnes of black mass with a representative composition of 18% Ni, 8% Co, 5% Mn, 3.5% Li, 2% Cu, and 3% Al. Incoming black mass is initially slurred with recycled process condensate. Concurrently, *Gluconobacter oxydans* is cultivated on glucose substrates within dual production fermenters, directly oxidizing glucose to gluconic acid at 30°C and pH 5.5. This upstream fermentation operates under hygienic rather than strictly sterile conditions: the accumulation of gluconic acid at pH 3 naturally inhibits competitive microbial contamination [13], allowing the plant to adopt lower-cost hygienic standards typical of industrial brewing rather than pharmaceutical-grade operations. Leaching is conducted across four batch reactors maintained at 55°C for a 30-hour residence time, aligning with optimized parameters established by Alipanah et al. [4], with ferrous sulfate added at 3.75-4.51 kg per kg of cobalt recovered. Maintaining a 15% free-acid excess requires a biolixiviant concentration near 900 mM, which directly dictates the volumetric throughput and scale of the upstream fermentation unit. Given that the organic broth is relatively dilute, with 150 g/L representing the upper limit reported in literature for this strain [12], the aqueous phase of the fermentation broth constitutes the vast majority of the leach fluid. As a result, the maximum pulp density is constrained to 4.2% by broth concentration limits rather than selected as an independent operational variable (Appendix G).

In the downstream separation circuit, dissolved copper is recovered via cementation onto scrap iron, leveraging metallic copper reduction behaviors analogous to established acidic recovery circuits [22]. Iron and aluminum are subsequently oxidized and precipitated at pH 5.5—a stage where PHREEQC chemical speciation modeling (Appendix F) confirms quantitative precipitation of iron starting from pH 3 and 91% precipitation efficiency for aluminum by pH 4.5 [19]. Three consecutive solvent extraction circuits follow: D2EHPA selectively extracts manganese near pH 3; Cyanex-272 separates cobalt from nickel across a distinct pH window, evaluated by Tait at pH_{50} values of 5.3 and 7.0 [15]; and Versatic-10 extracts nickel at pH 6.5. Recovered cobalt and nickel are crystallized as hydrated sulfate salts with mother liquors recycled back to the circuit, making internal bleed streams rather than single-pass efficiency the primary determinant of yield loss [17]. Lithium is subsequently precipitated directly as Li_3PO_4 , an extraction made viable by its low aqueous solubility from the 1.9 g/L barren stream without requiring thermal evaporation upstream. Remaining solution streams are concentrated for internal water recovery, leaving a single process bleed that is thermally dried into a crude sodium-gluconate technical salt, thereby achieving a zero liquid discharge (ZLD) footprint.

Under this base-case architecture, operating expenses exceed those of conventional hydrometallurgical facilities due to structural economic factors rather than operational inefficiencies. Total fixed capital expenditure is estimated at approximately \$118M, with annual operating costs ranging from \$33M to \$47M against baseline product revenue estimated at \$46M to \$58M (Appendix I). Revenue calculations are based on contained metal values derived from USGS commodity benchmarks [24], [25], noting that commercial pricing agencies do not index

Li_3PO_4 and battery-grade sulfate prices remain behind proprietary subscriptions [28]; actual commercial salt sales would likely realize higher margins. Also, glucose substrate consumption alone accounts for 53% of total operating expenses. At a fundamental chemical level, gluconic acid is a heavy, monoprotic acid derived from refined agricultural sugars, whereas sulfuric acid is a lightweight, diprotic molecule produced as a byproduct of industrial sulfur recovery. Expressed per unit of equivalent acidity delivered to the leaching reactor, gluconic acid costs \$80-130 per kilo-equivalent compared to just \$5-7 for sulfuric acid (Appendix I). This 11-to-27 fold cost disparity is further compounded by the low concentration of the biological broth, which imposes large fermenter volumes, high air compression loads, and significant evaporation requirements. This chemical cost premium, rather than the efficiency of the leaching reactions themselves, represents the primary economic barrier when competing against traditional hydrometallurgy.

Crucially, this cost disparity reflects present market conditions rather than an immutable limitation of biological processing. Three strategic process levers can significantly reduce this economic gap. First, replacing refined glucose syrup with unrefined corn stover hydrolysate, the low-cost carbon source assumed in original models by Alipanah et al. [4], reduces annual operating expenses by approximately \$14.5M. Second, recycling gluconate directly back to the leach circuit using bipolar-membrane electrodialysis (BMED) to form a closed acid loop saves an additional \$14.2M annually. Third, increasing fermentation broth titers yields negligible operating expense savings (roughly \$0.2M/yr), as reduced thermal evaporation requirements are minor relative to raw sugar costs; higher titers primarily reduce capital expenditure by enabling smaller equipment volumes. Implementing these process levers in tandem lowers annual operating costs to approximately \$20M, expanding net operating margins from \$8M-20M to \$27M-38M. Because these operational adjustments depend directly on feedstock selection and acid recycling configurations, today's cost differential should be interpreted as a mutable engineering challenge rather than an inherent flaw of the biohydrometallurgical pathway.

When evaluated alongside incumbent commercial technologies, the operational trade-offs are clearly defined. Traditional hydrometallurgical refining of black mass incurs processing costs of approximately \$1,500 to \$1,800 per tonne of feed material, whereas the unoptimized bioleaching flowsheet currently ranges from \$3,300 to \$4,700 per tonne prior to applying the proposed optimization levers. While bioleaching remains the more capital-intensive alternative today, it offers a fundamentally superior environmental profile rather than incremental efficiency gains. Conventional mineral-acid raffinate produces a persistent inorganic waste stream requiring ongoing long-term management, whereas gluconate-rich raffinate is fully biodegradable, featuring a theoretical oxygen demand of 0.90 g O_2 per gram that represents a treatable organic load [21]. Furthermore, the proposed zero liquid discharge design completely eliminates liquid wastewater effluents by drying combined bleed streams into saleable technical-grade salts; recovering condensate from dryer exhaust would fully close the water circuit against a baseline fresh water demand of 6.9 t/h. The reported overall CO_2 intensity for microbial bioleaching ranges between 16 and 19 kg per kg of recovered cobalt, compared to 43-91 kg for conventional

mineral acid routes [4]. Additionally, operating the leaching reactors at mild temperatures (55°C) avoids the high thermal energy requirements of pyrometallurgy while preventing the permanent loss of lithium and aluminum to slag phases.

Two additional operational benefits derive directly from the overall flowsheet architecture rather than the underlying leach chemistry. Recovering lithium via direct phosphate precipitation achieves a high overall recovery of 94.1% of feed lithium without requiring energy-intensive evaporation steps, outperforming conventional processing routes that frequently lose lithium to raffinate or slag streams. Furthermore, because gluconate functions as a natural chelating agent, the leach liquor retains metals in solution at pH ranges where iron and aluminum cleanly hydrolyze and precipitate, mitigating the formation of bulky ferric hydroxides that entrain and lose valuable cobalt. These technical advantages must be weighed against notable operational constraints. The required fermentation capacity exceeds typical industrial scales for citric or gluconic acid production, necessitating a multi-train fermenter installation. The facility consumes 4.8 t/h of refined glucose, introducing an agricultural supply reliance absent in conventional sulfuric acid plants, and the dilute biological broth approximately doubles thermal evaporator duty relative to high-titer bases. It is also important to note that the cumulative recovery figures reported here (80.3% Co, 78.4% Ni, 94.1% Li) reflect overall process yield from raw feed black mass to final bagged product, accounting for all downstream separation losses; these should not be directly equated with initial leaching extraction efficiencies (Appendix B), such as the 86% cobalt leaching yield reported by Alipanah et al. [4]. Ultimately, these findings support a clear conclusion: while microbial bioleaching is not yet cost-competitive with conventional processes, it offers immediate environmental advantages, with the remaining economic gap representing a solvable feedstock optimization and acid-recycle challenge.